Linear algebra and geometry 1 - TTPU

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Grade	**-1.05** out of 30.00 (**-4**%)

Question 1

Incorrect

Mark -15.00 out of 100.00

Let $\vec{i}, \vec{j}, \vec{k}$ the versors of the coordinate axes of $\mathbb{R}^3$. For every $a \in \mathbb{R}$, let us consider the vectors

$$\vec{u} = a\vec{j} + 2\vec{k}, \quad \vec{v} = -a\vec{i} + \vec{k}, \quad \vec{w} = -2\vec{i} - \vec{j}.$$

Which of the following statements is true?

- (a) There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar. ✗
- (b) For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly independent.
- (c) There exists a unique $a \in \mathbb{R}$ such that the vectors $\vec{u}, \vec{v}, \vec{w}$ are linearly independent.
- (d) For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.

Risposta errata.

The correct answer is: For every $a \in \mathbb{R}$ the vectors $\vec{u}, \vec{v}, \vec{w}$ are coplanar.

Question 2

Incorrect

Mark -15.00 out of 100.00

Given the plane $\pi : \ 3x - 5y + z = 0$ and the straight line $r : \begin{cases} 2x + y - z = 1 \\ 3x + 2y + z = 1. \end{cases}$, find the correct statement.

- (a) $r \cap \pi = \emptyset$. ✗
- (b) r and π are orthogonal.
- (c) $r \subseteq \pi$.
- (d) None of the other statements is correct.

Risposta errata.

The correct answer is: r and π are orthogonal.

Question 3

Incorrect

Mark -15.00 out of 100.00

Let us consider the linear system with 4 variables depending on the real parameter $k \in \mathbb{R}$:

$$\begin{cases} x - t = 0 \\ -x + 2y + 2z + t = 4 \\ y + z = k. \end{cases}.$$

Find the right statement.

- (a) For any $k \in \mathbb{R}$ the system does not admit any solution. ✗
- (b) If $k \neq 2$ the system admits only one solutions.
- (c) If $k = 2$ the system admits ∞^2 solutions.
- (d) If $k = 2$ the system admits ∞^1 solutions.

Risposta errata.

The correct answer is: If $k = 2$ the system admits ∞^2 solutions.

Question 4

Incorrect

Mark -15.00 out of 100.00

Let $\cdot$ denote the usual matrix multiplication. If

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 5 \\ 0 & 0 \end{bmatrix},$$

then

- (a) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$. ✗
- (b) $\det(\mathbf{A} \cdot \mathbf{B}) = 0$.
- (c) $\det(\mathbf{A} \cdot \mathbf{B}) = 5$.
- (d) $\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{B}) \cdot \det(\mathbf{A})$.

Risposta errata.

The correct answer is: $\det(\mathbf{A} \cdot \mathbf{B}) = 5$.

Question 5

Incorrect

Mark -15.00 out of 100.00

Let $\vec{w} = (1, 1, 1)$. Let $f : \mathbb{R}^3 \to \mathbb{R}^3$ be the endomorphism defined by:

$$\vec{v} \in \mathbb{R}^3 \to f(\vec{v}) = (\vec{v} \cdot \vec{w})\vec{w} + 2\vec{v}$$

Which one of the following statements is true?

- (a) $\vec{w}$ is not an eigenvector of f. ✗
- (b) f has no real eigenvalues.
- (c) None of the other statements is correct.
- (d) If $\vec{v} \neq \vec{0}$ is perpendicular to $\vec{w}$ then $\vec{v}$ is an eigenvector of f.

Risposta errata.

The correct answer is: If $\vec{v} \neq \vec{0}$ is perpendicular to $\vec{w}$ then $\vec{v}$ is an eigenvector of f.

Question 6

Incorrect

Mark -15.00 out of 100.00

In $\mathbb{R}^4$ let us consider the vector subspace

$$V = \{ (x, y, z, w) \mid x + y + z + w = 0, \ x - 3y = 0 \}.$$

Which one of the following statements is true?

- (a) V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(0, 0, 1, 1)$.
- (b) A basis of V is formed by 3 vectors.
- (c) V is a subspace of dimension 1.
- (d) V is the vector subspace generated by the vectors $(3, 1, 0, -4)$ e $(0, 0, 1, -1)$.

Risposta errata.

The correct answer is: V is the vector subspace generated by the vectors $(3, 1, 0, -4)$ e $(0, 0, 1, -1)$.

Question 7

Incorrect

Mark -15.00 out of 100.00

Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be the endomorphism $f(x, y) = (-x + 2y, -y)$.

Which one of the following statements is true?

- (a) f has two distinct eigenvalues.
- (b) $(0, 1)$ is an eigenvector.
- (c) $(2, 0)$ is an eigenvector.
- (d) f is simple.

Risposta errata.

The correct answer is: $(2, 0)$ is an eigenvector.

Question 8

Incorrect

Mark -15.00 out of 100.00

In the euclidean 2-dimensional space with a fixed cartesian system, let h be a real parameter and consider the conic $\mathcal{C}_h$ having equation

$$4hx^2 + hy^2 + 4(h+1)xy + 2x = 0.$$

Which of the following statements is true?

- (a) C_h is a hyperbola for every $h \in \mathbb{R}$. ✗
- (b) C_h is a parabola for every $h \in \mathbb{R}$.
- (c) C_h is degenerate for $h = 0$.
- (d) C_h is an ellipse for every $h \in \mathbb{R}$.

Risposta errata.

The correct answer is: C_h is degenerate for $h = 0$.

Question 9

Correct

Mark 100.00 out of 100.00

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the circumferences of equations

$$\mathcal{C}_1 : x^2 + y^2 + z^2 - 4 = y = 0, \quad \mathcal{C}_2 : x^2 + y^2 + z^2 - 4 = y - 1 = 0,$$

having radius respectively R_1 and R_2.

Which of the following statements is true?

- (a) $R_1 < R_2$.
- (b) $R_1 > R_2$. ✓
- (c) $\mathcal{C}_1$ consists of only one point.
- (d) $R_1 = R_2$.

Risposta corretta.

The correct answer is: $R_1 > R_2$.

Question 10

Incorrect

Mark -15.00 out of 100.00

Let us consider the quadratic form defined by

$$q(x,y) = (x,y)A\begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

- (a) $q(x,y) = 0$ is the equation of two lines passing through the origin.
- (b) The matrix A has two eigenvalues of different sign. ✗
- (c) $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$.
- (d) The quadratic form $q(x,y)$ is negative definite.

Risposta errata.

The correct answer is: $q(a,b) > 0$ for every $(a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$.